Junsu Kim

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RESEARCH INTERESTS

Computer Architecture | Systems for ML | ML for Systems

- **Efficient LLM Inference**
  - **Speculative Decoding:** Designing efficient disaggregated drafting and self-drafting schemes that scale with speculation length and batch size, and that generalize across recent AI models, including MoE and linear-attention architectures.
  - **KV Cache Management:** Developing high-fidelity KV cache compression and tiered KV cache management that scales to long-context and long-generation serving while maintaining accuracy.
- **Memory and Storage Systems for AI**
  - **GPU Tiered Memory Management:** Designing kernel scheduling and memory management for heterogeneous GPU memory systems backed by CXL and SSDs to overcome the GPU memory wall with minimal performance loss.

EDUCATION

The University of British Columbia | Vancouver, BC, Canada — Sep. 2025 – Current
Ph.D. in Electrical and Computer Engineering • Advisor: Prof. Prashant Nair • GPA: 4.0/4.0

Korea University | Seoul, Korea — Sep. 2023 – Aug. 2025
M.S. in Electrical Engineering • Advisor: Prof. Yunho Oh • GPA: 4.0/4.0

Hanyang University | Seoul, Korea — Mar. 2014 – Feb. 2021
B.S. in Electronic Engineering • Advisor: Prof. Ki-Seok Chung • GPA: 3.81/4.0 (*Summa Cum Laude*)

SELECTED PUBLICATIONS

Conference [C] and Journal [J] Papers (* denotes co-first author)

[C6] Jeonghyun Woo*, **Junsu Kim***, Aamer Jaleel, and Prashant Nair, "Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense," *The 53rd International Symposium on Computer Architecture*. ISCA 2026

[J2] **Junsu Kim**, Jaebeom Jeon, Jaeyong Park, Sangun Choi, Minseong Gil, Seokin Hong, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "Memory Oversubscription-Aware Tensor Migration Scheduling for GPU Unified Storage Architecture," *IEEE Computer Architecture Letters*. CAL 2025

[C5] **Junsu Kim** and Suhyun Kim, "Salient Frequency-aware Exemplar Compression for Resource-constrained Online Continual Learning," *The 39th Annual AAAI Conference on Artificial Intelligence*. AAAI 2025

[C2] Jaebeom Jeon, Minsung Gil, **Junsu Kim**, Jaeyoung Park, Gunjae Koo, Myung-Kuk Yoon, and Yunho Oh, "VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing," *The 53rd International Conference on Parallel Processing*. ICPP 2024

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung, "HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer," *The 39th IEEE International Conference on Computer Design*. ICCD 2021

Preprints (project names only)

[P1] **First author** — "Scalable Disaggregated Speculative Decoding"

WORK EXPERIENCE

d-Matrix | Santa Clara, CA, United States — Jun. 2026 – Sep. 2026
AI Systems Intern (Remote) — Manager: Nithesh Kurella • Mentor: Nikhil Pratap Ghanathe

The University of British Columbia | Vancouver, BC, Canada — Sep. 2025 – Current
Research Assistant, Systems and Architectures (STAR) Lab — Advisor: Prof. Prashant Nair

Korea University | Seoul, Korea — Sep. 2023 – Aug. 2025
Research Assistant, Computer Architecture and System Software Lab (ComSys) — Advisor: Prof. Yunho Oh

Korea Institute of Science and Technology | Seoul, Korea — May. 2022 – Aug. 2023
Research Assistant, Korea Data Science Team (KDST) — Supervisor: Dr. Suhyun Kim

Hanyang University | Seoul, Korea — Dec. 2019 – Mar. 2020, Aug. 2020 – Nov. 2020
Research Assistant, Embedded System on Chip Laboratory (ESOC Lab) — Advisor: Prof. Ki-Seok Chung
Research Assistant, Computer Architecture and System SW Lab (CASS Lab) — Advisor: Prof. Yongjun Park

TEACHING EXPERIENCE

The University of British Columbia | Vancouver, BC, Canada — Fall 2025
Teaching Assistant — Computer Architecture (CPEN 411)

Korea University | Seoul, Korea — Spring 2024, Fall 2024
Teaching Assistant — Computer Architecture

School for the Blind | Chuncheon, Korea — Mar. 2017 – Feb. 2019
Assistant Teacher (Alternative Military Service)

HONORS AND AWARDS

- ISCA Student Travel Grant → **$1,000 USD** — May. 2026
- Faculty of Applied Science Graduate Award, University of British Columbia → **$7,200 CAD** — Nov. 2025
- AAAI Student Travel Grant → **$1,600 USD** — Mar. 2025
- Korea University Graduate School Scholarship → ∼**$5,000 CAD/semester** — Spring 2024, Fall 2024
- Korea University Graduate School Scholarship for Outstanding New Student → ∼**$5,000 CAD** — Fall 2023
- Hanyang University Scholarship → ∼**$5,000 CAD/semester** — 2014 – 2019

RESEARCH PROJECTS

Scalable Disaggregated Speculative Decoding
Advisor: Prof. Prashant Nair, The University of British Columbia • Collaborator: d-Matrix — Mar. 2026 – Current

- To be updated

Accuracy-preserving Tiered Key-Value Cache Management for Large Language Models
Advisor: Prof. Prashant Nair, The University of British Columbia — Sep. 2025 – Mar. 2026

- Proposed a tiered KV cache management framework that leverages a full KV cache stored in host DRAM and a compressed KV cache residing in GPU HBM
- Proposed reforwarding that rejects a new output token generated by the compressed KV cache and regenerates it with the full KV cache depending on output token confidence
- Proposed KV cache recomputation that recomputes the KV cache generated by the compressed KV cache during reforwarding
- Reduced KV cache size by **60%** while achieving an average speedup of **65.4%** on LongWriter (long-generation) and LongBench-v1 (long-context), with less than 1% accuracy loss
- *Contributions:* first author — motivation study, idea, implementation

Hardware Supports for Enabling Arbitrary Numeric Format on GPUs
Advisor: Prof. Yunho Oh, Korea University — May. 2024 – Nov. 2024

- Observed that GPUs support a limited set of numeric formats, wasting register files when processing arbitrary numeric formats
- Employed a bitslice representation that transposes data elements, packing arbitrary numeric formats without register wastage
- Proposed a Bitslice Vector multiplier and adder, constructing a tree structure to replace the multiplication-adder tree in a Tensor core
- *Contributions:* co-author — motivation study, idea, implementation, paper write-up

A Behavioral Analysis of CXL Memory Systems
Advisor: Prof. Yunho Oh, Korea University • Collaborator: SK hynix — Sep. 2023 – Sep. 2024

- Observed the behavior of a real CXL-based system on datacenter and AI workloads in a CXL-based platform
- Analyzed how different promotion and demotion methods for CXL devices affected workload performance
- Presented performance modeling for datacenter workloads using different system factors (e.g., memory bandwidth, memory latency)
- *Contributions:* co-author — experiments, analysis, paper write-up

SKILLS

Languages & Tools: C/C++, Python, TensorFlow, PyTorch, Git, Verilog, Shell script

FULL PUBLICATIONS

Conference [C] and Journal [J] Papers (* denotes co-first author)

[C6] Jeonghyun Woo*, **Junsu Kim***, Aamer Jaleel, and Prashant Nair, "Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense," *The 53rd International Symposium on Computer Architecture*. **ISCA 2026**

[J2] **Junsu Kim**, Jaebeom Jeon, Jaeyong Park, Sangun Choi, Minseong Gil, Seokin Hong, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "Memory Oversubscription-Aware Tensor Migration Scheduling for GPU Unified Storage Architecture," *IEEE Computer Architecture Letters*. **CAL 2025**

[C5] **Junsu Kim** and Suhyun Kim, "Salient Frequency-aware Exemplar Compression for Resource-constrained Online Continual Learning," *The 39th Annual AAAI Conference on Artificial Intelligence*. **AAAI 2025**

[C4] Seondeok Kim*, Sangun Choi*, Jaebeom Jeon, **Junsu Kim**, Minseong Gil, Jaehyeok Ryu, and Yunho Oh, "Kubism: Disassembling and Reassembling K-Means Clustering for Mobile Heterogeneous Platforms," *The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems*.

[C3] Dongwon Yang, Jaebeom Jeon, Minseong Gil, **Junsu Kim**, Seondeok Kim, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "SSFFT: Energy-Efficient Selective Scaling for Fast Fourier Transform in Embedded GPUs," *The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems*.

[J1] Minseong Gil, Jaebeom Jeon, **Junsu Kim**, Sangun Choi, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs," *IEEE Embedded Systems Letters*.

[C2] Jaebeom Jeon, Minsung Gil, **Junsu Kim**, Jaeyoung Park, Gunjae Koo, Myung-Kuk Yoon, and Yunho Oh, "VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing," *The 53rd International Conference on Parallel Processing*. **ICPP 2024**

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung, "HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer," *The 39th IEEE International Conference on Computer Design*. **ICCD 2021**

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[P1] **First author** — "Scalable Disaggregated Speculative Decoding"

[P2] Co-author — "Hardware Supports for Enabling Arbitrary Numeric Format on GPUs"